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1 Appendix E: Spectral Unmixing and Classification

1.1 Objective

Unmix composite spectra and evaluate small, deterministic classification baselines on CHC features.

1.2 Planned Methods (src)

- `src/spectral_unmixing.py`
 - `nmf_unmix(spectra, n_components, seed=42)`
 - `lda_baseline(features, labels, seed=42)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_spectral_unmixing.py`
- Data: `output/data/spectral_unmixing.npz`
- Figure: `output/figures/spectral_unmixing_components.png`

1.4 Figure

1.5 Equation References

- Spectral overlap: see (??) analogs for information metrics; overlap in main text.

1.6 Reproducibility

- Run: `python3 scripts/generate_spectral_unmixing.py`
- Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??

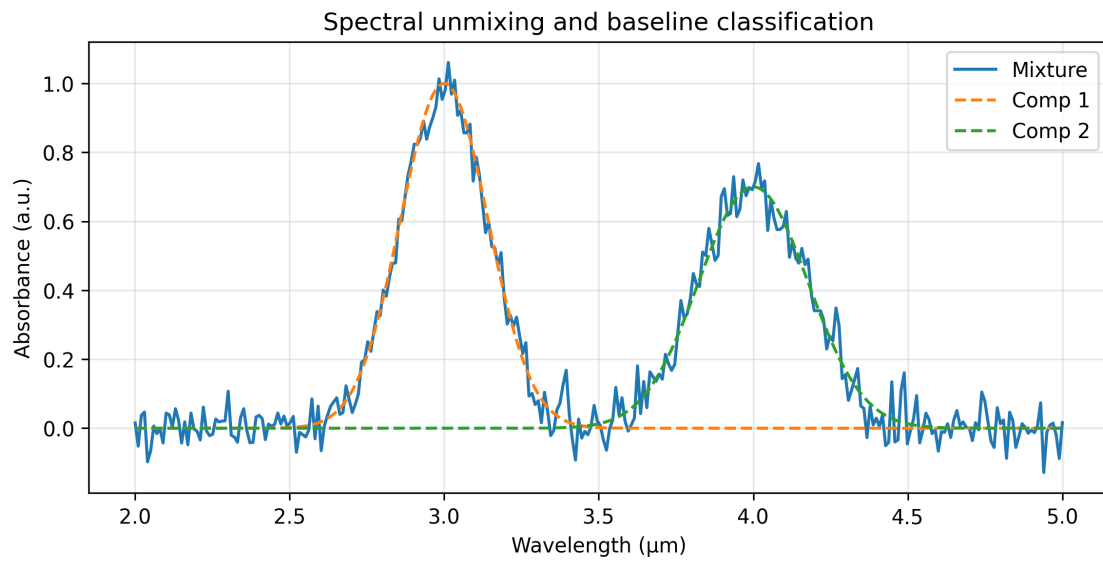


Figure 1: *Unmixed component spectra and reconstruction error across seeds; baseline classification accuracy on deterministic folds.*